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BSc in Electrical and Computer Engineering

# 5G NETWORKS: DEVELOPMENT OF A 5G CORE AND RADIO ACCESS NETWORK

PRE-DISSERTATION INFORMATION

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING

NOVA University Lisbon

*Draft: April 21, 2026*

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## ACRONYMS

|               |                                          |
|---------------|------------------------------------------|
| <b>COTS</b>   | Commercial Off-The-Shelf ( <i>p. 3</i> ) |
| <b>eNodeB</b> | Evolved Node B ( <i>p. 3</i> )           |
| <b>EPC</b>    | Evolved Packet Core ( <i>pp. 2, 3</i> )  |
| <b>RAN</b>    | Radio Access Network ( <i>pp. 2, 3</i> ) |
| <b>SDR</b>    | Software-Defined Radio ( <i>p. 3</i> )   |
| <b>UE</b>     | User Equipment ( <i>p. 3</i> )           |



# 4G NETWORK SETUP

Before developing the 5G network testbed, a 4G network was first established to serve as a performance baseline and a proof-of-concept for all hardware and software components. The 4G network setup was essential for several reasons:

1. Validating that the chosen hardware and software stack functioned reliably
2. Establishing a known and well-documented reference point for network behavior
3. Identifying potential deployment hurdles before scaling to 5G complexity

This appendix documents the complete 4G testbed architecture, deployment procedures, and validation results.

The network stack consists of two dedicated machines running the backbone infrastructure: one running Open5GS for the Evolved Packet Core (EPC), while the other runs srsRAN 4G for the Radio Access Network (RAN). Readers can use this section as a guide for replicating the network, and as a foundation for understanding the experimental setup that enabled the 5G network development described in the main thesis.

This section will provide a comprehensive guide on how the 4G network was established, including a description of the equipment used for the setup and the software chosen for the network stack, the necessary setup steps and specific configurations applied to optimize performance, and the testing methodologies employed to validate the network's functionality and performance.

As an effort to make the network as reproducible as possible, the setup process will be detailed step-by-step, starting from the initial hardware assembly to the final testing phase. This will include instructions on how to connect the various components, configure the network settings, and troubleshoot common issues that may arise during the setup process.

## .1 System Architecture Overview

The 4G testbed is a self-contained LTE system comprising three main functional domains:



- **RAN:** Handles air-interface communication using srsRAN Evolved Node B (eNodeB) software running on a dedicated machine, along with a Nuand BladeRF Software-Defined Radio (SDR) and appropriate antennas for signal transmission and reception
- **EPC:** Implements the core network functions using Open5GS software running on a separate machine, providing essential services such as authentication, mobility management, and data routing to the internet
- **User Equipment (UE):** Physical Commercial Off-The-Shelf (COTS) device configured to connect to the RAN and access the services provided by the EPC

Figure .1 presents the logical architecture.

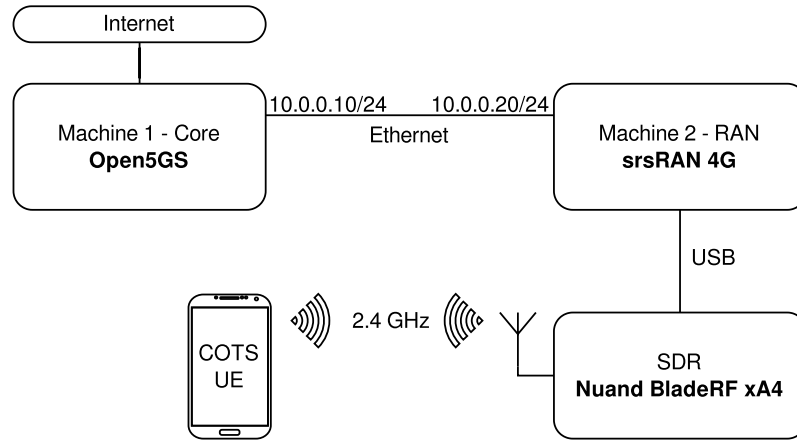


Figure .1: Logical architecture of the 4G network, illustrating the interactions between the RAN, EPC, and UE components.

## .2 Hardware Components

The hardware used for the 4G network setup is composed of two identical machines, a Software-Defined Radio and antennas.

The computers running the 4G network are DELL Optiplex Micro 7020, with the following specifications:

| Component        | Machine 1            | Machine 2            |
|------------------|----------------------|----------------------|
| CPU              | Intel Core i7-14700T | Intel Core i7-14700T |
| RAM              | 16 GB DDR5           | 16 GB DDR5           |
| Operating System | Ubuntu 22.04 LTS     | Ubuntu 22.04 LTS     |
| Storage          | 512 GB SSD           | 512 GB SSD           |

Table .2: Specifications of the machines used for the 4G network setup.

Data flows between the machines via a dedicated Ethernet backhaul network. The RAN on Machine 2 establishes a connection to the EPC on Machine 1 using SCTP

(Stream Control Transmission Protocol). When a UE initiates a connection, the RAN forwards authentication requests to the core, which validates the subscriber and allocates IP resources. All user data packets then traverse through the core network functions before reaching their destination.